

ELECTRO CHEMISTRY

- Which statement is correct
 - a galvanic cell is an electro chemical cell that converts the chemical energy of a spontaneous redox reaction into electrical energy.
 - If an external opposite potential is applied on daniel cell is more than 1.1V than it function as an electrolytic cell.
 - In a galvanic cell, the half cell in which oxidation takes place is called anode and it has a negative potential with respect to the solution.
 - All are correct
- Given the standard electrode potential
 $K^+/K = -2.93V$, $Ag^+/Ag = 0.80V$
 $Hg^{2+}/Hg = 0.79V$; $Mg^{2+}/Mg = -2.37V$,
 $Cr^{3+}/Cr = -0.74V$
 Correct order of their reducing power is –
 - $Ag > Hg > Cr > Mg > K$
 - $K > Mg > Cr > Hg > Ag$
 - $K > Cr > Mg > Ag > Hg$
 - $Cr > Mg > K > Hg > Ag$
- The conductivity of Electrolytic solutions depends on –
 - Size of the ions produced and their solvation
 - The nature of the solvent and its viscosity
 - Temperature
 - All of the above
- Which statement is incorrect about cell constant
 - Cell constant depends on the distance between the electrodes and their area of cross section
 - The dimension of cell constant is $length^{-1}$
 - The cell constant $G^* = R\kappa$
 - $G^* = \kappa/R$
- How much electricity in term of Faraday is required for the reducing 1mol of $KMnO_4$ to MnO_2
 - 3F
 - 5F
 - 1F
 - 2F
- The reaction which take place at cathode in Leclanche cell is
 - $HgO + H_2O + 2e^- \rightarrow Hg + 2OH^-$
 - $Zn \rightarrow Zn^{2+} + 2e^-$
 - $MnO_2 + NH_4^+ + e^- \rightarrow MnO(OH) + NH_3$
 - $PbO_2 + SO_4^{2-} + 4H^+ + 2e^- \rightarrow PbSO_4 + 2H_2O$
- Effeciency of fuel cell to produce electricity is approximately
 - 40%
 - 50%
 - 60%
 - 70%
- Example of corrosion is –
 - Rusting of iron
 - Tarnishing of silver
 - development of green coating on copper and bronze
 - All of the above
- At 298K, K_c for Daniel cell is approximately ($E^\circ = 1.10V$)
 - 2×10^{37}
 - 2×10^{38}
 - 4×10^{37}
 - 4×10^{39}
- The volume of hydrogen gas liberated at STP when a current of 5.36 ampere is passed through dil. H_2SO_4 for 5 hours will be
 - 5.6L
 - 11.2L
 - 16.8L
 - 22.4L
- Which one of the following statement is incorrect regarding the galvanic cell in which the reaction.
 $Zn(s) + 2Ag^+(aq) \rightarrow Zn^{2+}(aq) + 2Ag(s)$ takes place
 - Zn is negatively charged electrode
 - Ag is negatively charged electrode
 - The flow of current is partly due to flow of electrons and partly due to flow of ions
 - Current flow from Ag to Zn in external circuit
- The electrolysis of silver nitrate solution is carried out using silver electrods. Which the following reaction occurs at the anode?
 - $Ag \rightarrow Ag^+ + e^-$
 - $Ag^+ + e^- \rightarrow Ag$
 - $2H_2O \rightarrow 4H^+ + O_2 + 4e^-$
 - $4OH^- \rightarrow 2H_2 + O_2 + 4e^-$
- The equilibrium constant (K) for the reaction
 $Fe^{2+}(aq) + Ag^+(aq) \rightarrow Fe^{3+}(aq) + Ag(s)$ will be
 Given $E^\circ(Fe^{3+}/Fe^{2+}) = 0.77V$; $E^\circ(Ag^+/Ag) = 0.80V$
 - 100
 - 3.16
 - 10
 - 0.5

- 14.** The EMF of the cell
 $\text{Pt(s)}/\text{Br}_2(\ell)/\text{Br}^-(0.010\text{M})\parallel\text{H}^+(0.010\text{M})/\text{H}_2(\text{g})$
 (1 bar)/Pt(s) at 298 K is
 (Given $E^\circ_{\text{Br}_2/\text{Br}^-} = 1.09\text{V}$)
 (1) 1.09V (2) 1.21V
 (3) -1.326V (4) 0.854 V
- 15.** The emf of Daniel cell is 1.1Volt. If the value of Faraday constant is 96500 C per mol, the change in free energy in KJ is
 (1) 212.30 (2) -212.30
 (3) 106.15 (4) -106.15
- 16.** The ionization constant of a weak electrolyte is 25×10^{-6} while the equivalent conductance of its 0.01M solution is $19.6 \text{ Scm}^2 \text{ eq}^{-1}$. The equivalent conductance of the electrolyte at infinite dilution (in $\text{Scm}^2 \text{ eq}^{-1}$) will be
 (1) 250 (2) 196 (3) 392 (4) 384
- 17.** The molar ionic conductances at infinite dilution of K^+ and SO_4^{2-} are 73.5 and $160 \text{ Scm}^2 \text{ mol}^{-1}$ respectively. The molar conductance of solution of K_2SO_4 at infinite dilution will be
 (1) 233.5 (2) 307 (3) 153.5 (4) 467
- 18.** Iron can be protected by coating with zinc or tin. If coating is broken.
 (1) iron will corrode faster if coated with zinc.
 (2) Iron will corrode faster if coated with tin.
 (3) iron will corrode faster in both cases
 (4) iron will not undergo any corrosion in boths case
- 19.** I_2 and Br_2 are added to a solution conataining 1M each of I^- and Br^- ions. Which of the following reaction will take place? (Given : Standard reduction potentials of I_2 and Br_2 are 0.53 and 1.09 Volts repectively)
 (1) Iodine will reduce bromide ions
 (2) Bromine will reduce iodide ions
 (3) Iodide ions will reduce bromine
 (4) Bromide ion will reduce iodine